

End-to-End Sentiment Analysis System Using NLP & Machine Learning

1. Introduction

Sentiment analysis is a Natural Language Processing (NLP) task used to determine whether a piece of text expresses positive or negative sentiment. The objective of this project was to build a complete sentiment analysis system capable of classifying movie reviews using machine learning techniques.

2. Dataset Description

The IMDb Movie Reviews dataset was used for this project.

Characteristics:

- * Approximately 50,000 movie reviews
- * Binary sentiment classification
- * Positive and negative labels
- * Balanced dataset

Target Variable:

- * Positive (1)
- * Negative (0)

3. Data Exploration and Analysis

The dataset was explored to understand:

- * Class distribution
- * Missing values
- * Duplicate records
- * Review lengths

Visualizations created:

- * Sentiment distribution chart
- * Review length distribution
- * Word cloud
- * Word frequency analysis

Key Findings:

- * Dataset was balanced between positive and negative reviews.
- * Review lengths varied significantly.
- * Frequently occurring words reflected sentiment polarity.

4. Text Preprocessing

The following preprocessing steps were applied:

- * Lowercasing
- * Tokenization
- * Stopword removal
- * Punctuation removal
- * Lemmatization

These steps improved text quality and reduced noise.

5. Feature Engineering

Two feature extraction techniques were implemented:

Bag of Words (BoW)

Converts text into numerical vectors based on word frequencies.

TF-IDF

Assigns importance scores to words while reducing the impact of common words.

TF-IDF was selected as the primary feature representation due to stronger performance.

6. Machine Learning Models

Three machine learning models were trained:

Logistic Regression

A linear classification model widely used for text classification.

Naive Bayes

A probabilistic classifier effective for NLP tasks.

Random Forest

An ensemble learning method based on multiple decision trees.

7. Model Evaluation

Evaluation metrics used:

- * Accuracy
- * Precision
- * Recall
- * F1 Score
- * Confusion Matrix
- * Cross Validation

Results:

Model	Accuracy
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Logistic Regression	88.42%
Naive Bayes	85.32%
Random Forest	85.00%

Cross Validation Accuracy:

86.24%

8. Hyperparameter Tuning

GridSearchCV was used to optimize Logistic Regression.

Best Parameter:

C = 1

The optimized model produced the highest validation performance.

9. Final Model Selection

Logistic Regression was selected as the final model because it achieved the highest overall performance while maintaining interpretability and computational efficiency.

Final Metrics:

- * Accuracy: 88.42%
- * Precision: 87.34%
- * Recall: 89.70%
- * F1 Score: 88.51%

* Cross Validation Accuracy: 86.24%

10. Inference Pipeline

A prediction pipeline was developed to classify new movie reviews.

Example:

Input:

"This movie was fantastic and enjoyable."

Output:

Positive

Input:

"This movie was terrible and boring."

Output:

Negative

11. Conclusion

This project successfully implemented a complete sentiment analysis system using NLP and Machine Learning techniques. After comparing multiple models, Logistic Regression achieved the best performance and was selected as the final model. The resulting system can effectively classify movie reviews into positive and negative sentiment categories.